

BT 302 (IMMUNOLOGY)

FINAL TERM CURRENT FILE 2024

SOLVED & COMPOSED BY

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OBJECTIVE

- 1) **Immunology** is a study of **the immune system, its functions, and disorders.**
- 2) Global occurrence infection ... **Pandemic**
- 3) C4a ,C3a,C5a are **Potent anaphylatoxin**
- 4) ADCC another form... **Type II hypersensitivity**
- 5) Immune response independent of antibody called.... **Cell mediated immunity**
- 6) Toxin production is neutralized by antibodies name (**antitoxins**)
- 7) Induction of active humoral immunity___ **Passive Immunization**
- 8) Adjuvant have **two Types**
- 9) components form immunoglobulin deficiency is.... **IgA deficiency**
- 10) IL-12 is cytokine of.... **Inflammation**
- 11) pneumococcus polysaccharide serve ... **T independent Antigen**
- 12) Alpha -Feto protein level increasing during ___ **hepatocellular carcinoma**
- 13) Molecule mass ranges that tends to make less substances poor immunogens.
- 14) Primary role of IgE___ **Involved in allergic reactions**
- 15) classic pathways for complement activation initiate also called ___ **antibody dependent complement activation.**
- 16) AIDS caused by(**HIV**)
- 17) acute rejection are....(**1st set of rejection**)
- 18) isograft between individual of same genetic makeup ___ **identical twin.**
- 19) tolerance to ..(**t cell**).... is longer as compared to b-cell.
- 20) Animals with hybrid nature of Immune cells with host and donor cells are___ **Chimeras**
- 21) Patient suffering AIDS are susceptible to___ **Kaposi Sarcoma**
- 22) B-cells are activated directly without the help of T-cells for antibodies production in___ **T- independent antigens**
- 23) Infectious diseases that spread from person to person are said to be___ **Contagious**

- 24) T- Dependent antigens contain ----- type of epitopes with few copies___ **Same**
- 25) Autoimmune disorders___ **abnormal response against self-tissues**
- 26) Type 1 hypersensitivity is mediated by___ **IgE**
- 27) Pro- inflammatory cytokines are involved in ----- of inflammatory process.
____ **Up regulation**
- 28) These all are true for immune evasion EXCEPT -----**maximize the ability of organisms to develop infection**
- 29) These all are true for inflammation EXCEPT ---- **Expressed in myeloid cells**
- 30) **Adjuvant** are of ____ **2 types**
- 31) Lymph nodes, spleen and Peyer's patches are. _____ **Secondary lymphoid organs**
- 32) Second most common serum antibody... **IGA**
- 33) Humans can make 10^7 - 10^8 different Ab molecules
T-cells maturation in... **Thymus**
- 34) Class switching of immunoglobulin occurs from IgM to ----- **IgG**
- 35) Cytidine deamine is present in ----- **B cells**
- 36) The gene for ----- chain is located on **chromosome 14** _____ **Heavy chain**
- 37) MHC II molecules contain highly conserved ----- regions for binding to CD4 receptors. _____ **$\alpha 2$ & $\beta 2$**
- 38) Antigen binding Groove of Class II MHC can accommodate the peptides consisting of _____ **13-25**
Avidity is influenced by the ----- of both antigen and antibody. **Valency**
- 39) Reaction of type 1 hypersensitivity _____ **Usually 15-30 min, sometimes 10-12 hours**
- 40) Time required for type I _____ **3-10 hours**
- 41) Th1 cells release cytokine like IFN γ upon activation which is inhibitory for -----
Th2 cells
- 42) MHC stands for ____ **Major Histocompatibility Complex.**
- 43) Symptoms of Inflammation _____ **pain, heat, redness, swelling & loss of tissue functions**
- 44) Exudation of plasma proteins & fluids _____ **Edema**
- 45) Which statement is true regarding t-cell and b-cell _____ **t-cell matures in thymus and b cell matures in bone marrow**
- 46) Soluble antigens are _____ **Less immunogenic.**
- 47) Framework Regions are Products of differentregion genes.
_____ **variable**
- 48) Least common serum antibody is _____ **IgE**

- 49) Affinity.....**Strength of binding B/W single antigenic determinant & single antibody site.**
- 50) Agglutination.....**Particulate nature of antigen causes the reaction of antibody to form visible clumps.**
- 51) Unknown Ab can be determined by_____ **direct ELISA.**
- 52) Role of IgE in allergic reactions__**Involved in allergic reactions (Anaphylaxis)**
- 53) Primary function of coagulation__ **Antimicrobial activity**
- 54) How many kinds of polyadenylation__ **two**
- 55) Example of live attenuated vaccine__ **Bacille Calmette-Guerin vaccine (BCG)**
- 56) Time span of acute rejection__ **1-3 weeks**
- 57) ADCC is another form of__ **Antibody dependent Cellular Cytotoxicity**
- 58) MASP stands for ____ **Mannose-Associated Serine Proteases**

-----**SUBJECTIVE**-----

1. Active immunization vaccines?

It is the induction of immunity after exposure of host to antigen. In this case, antigenic exposure is mandatory. Active immunization can occur naturally by exposing to microbe or other antigen when there is no prior exposure before the entry of antigen. Similarly there is also no pre made antibodies in the host's serum. In this form of immunization, immune system develops antibodies against the microbe after its exposure. This process of antibodies generation is considered slow. Moreover a memory response is also generated in which antibodies remained in use for longer time. This form of immunization can be artificial in nature which is achieved by either injection of microbe before natural exposure or treated microbes or toxins

2. Opsonization?

It is the process by which phagocytosis of microbes by immune cells get enhanced. Those proteins which undertake this process are called as opsonins. For example, some complement proteins like C3b, iC3b and C4b are good opsonins. These proteins attach to microorganism first making a complex. This complex of complement protein & microbe bind with complement receptor on the surface of phagocytic cell as a result phagocytosis of microbe occur by phagocytes

3. Structure of TCR?

The T-cell receptor (TCR) is a complex structure on T cells that recognizes antigens. It consists of:

1. **Alpha (α) and Beta (β) Chains:**
 - **Variable Regions ($V\alpha$ and $V\beta$):** Determine antigen specificity.
 - **Constant Regions ($C\alpha$ and $C\beta$):** Provide structural support.
 - **Transmembrane Regions:** Anchor the TCR in the cell membrane.
2. **CD3 Complex:** Associated subunits (CD3 ϵ , CD3 γ , CD3 δ , and CD3 ζ) crucial for signal transduction.
3. **Antigen Recognition:** TCRs recognize antigens presented by Major Histocompatibility Complex (MHC) molecules:
 - **MHC Class I:** Recognized by CD8⁺ T cells.
 - **MHC Class II:** Recognized by CD4⁺ T cells.
4. **Co-receptors:**
 - **CD4:** Binds to MHC Class II.
 - **CD8:** Binds to MHC Class I.

This structure enables T cells to recognize and respond to specific antigens, playing a crucial role in the adaptive immune system.

4. Characteristics of immunoglobulin?

Immunoglobulins (Ig) are antibodies produced by plasma cells with these key characteristics:

1. **Structure:**
 - Composed of two heavy (H) and two light (L) chains.
 - Variable (V) regions for antigen binding.
 - Constant (C) regions for effector functions.
 - Y-shaped molecule held by disulfide bonds.
2. **Classes:**
 - **IgG:** Most abundant, found in blood and extracellular fluid.
 - **IgA:** Found in mucosal areas and secretions.
 - **IgM:** First antibody produced, primarily in blood.
 - **IgE:** Involved in allergic reactions and defense against parasites.
 - **IgD:** Present on B cell surfaces and in small amounts in blood.
3. **Functions:**
 - **Antigen Binding:** Specific recognition of antigens.
 - **Neutralization:** Inactivates pathogens/toxins.

- **Opsonization:** Marks pathogens for phagocytosis.
 - **Complement Activation:** Triggers pathogen lysis.
 - **Agglutination:** Clumps pathogens for easier targeting.
 - **ADCC:** Targets infected cells for destruction by NK cells.
4. **Production:**
- **Primary Response:** Initial production, mainly IgM followed by IgG.
 - **Secondary Response:** Faster, robust IgG production due to memory B cells.

These properties allow immunoglobulins to effectively identify and neutralize foreign invaders.

5. **Parts of secondary lymphoid system?**

The secondary lymphoid system includes structures where lymphocytes become activated and initiate adaptive immune responses. Key parts of this system are:

1. **Lymph Nodes:**
 - Filter lymph fluid.
 - Sites for antigen presentation and lymphocyte activation.
2. **Spleen:**
 - Filters blood.
 - Red pulp: Removes old red blood cells.
 - White pulp: Immune responses to blood-borne antigens.
3. **Tonsils:**
 - Located at the throat and nasal passages.
 - First line of defense against inhaled or ingested pathogens.
4. **Appendix:**
 - Contains lymphoid tissue.
 - Plays a role in gut immunity.

6. **Complement proteins?**

Complement system is comprised of heat labile serum proteins which are able to destroy or lyse pathogens. This form of host defense is termed as one way of host defense against potential pathogens. These proteins are produced by variety of body's cells including hepatocytes (Liver cells), macrophages & gut epithelial cells. Among complement proteins, some proteins has the ability to bind with immunoglobulin molecules while some are involved in binding on the membrane component of various immune cells.

7. Endogenous & exogenous antigen difference?

Endogenous and exogenous antigens are two main types of antigens that the immune system recognizes and responds to:

Endogenous Antigens

- **Origin:** Generated within the body's cells.
- Produced within the cell, presented by MHC Class I, and targeted by CD8+ T cells.
- **Examples:** Viral proteins produced inside infected cells, tumor antigens from cancerous cells, and normal cellular proteins presented during routine maintenance.

Exogenous Antigens

- **Origin:** Enter the body from the external environment.
- presented by MHC Class II, and targeted by CD4+ T cells.
- **Examples:** Bacteria, fungi, parasites, pollen, and toxins.

8. Activation of B-cell receptor?

B-cells are activated when there is high concentration of antigen for example in case of bacteria the lipopolysaccharides antigens exist in greater quantity. B-cell activation occurs after binding of antigen to a surface molecule which is known as B-cell receptor. Afterward there is bypassing of the early part of the biochemical pathway mediated by specific antigen receptor. Cross linking of immunoglobulin receptor occurs due to repeating determinants on antigenic structure. As a result the transformation of B-cells occurs into plasma cell for immunoglobulin secretion.

9. Active / passive immunization?

Active immunization: It is the induction of immunity after exposure of host to antigen. In this case, antigenic exposure is mandatory. Active immunization can occur naturally by exposing to microbe or other antigen when there is no prior exposure before the entry of antigen. In this form of immunization, immune system develops antibodies against the microbe after its exposure. This process of antibodies generation is considered slow.

Passive immunization: It is the induction of active humoral immunity. In this kind of immunization, antibodies are mandatory. Passive immunization can occur naturally by transfer of maternal antibodies to fetus through placenta. However, the artificial induction can be done by injecting gamma globulins from other individuals or animals to host.

10. Adaptive immune system?

- Response is antigen-dependent
- There is a lag time between exposure and maximal response
- Antigen-specific
- Exposure results in immunologic memory

Hypersensitivity refers to excessive, undesirable (damaging, discomfort-producing and sometimes fatal) reactions produced by the normal immune system. It is the state of undesirable reaction by normal immune system, in which immune system overreacts against various stimuli.

11. Hypersensitivity and its types?

In **hypersensitivity**, the protective role of immune system become harmful for its host. This condition is also termed as Allergy which is an abnormal response against the otherwise harmless environment stimulus e.g food, pollen & animal dander.

Followings are the important types of hypersensitivity reactions

Type I = (Also known as Immediate or Anaphylactic reactions, Involve various tissues)

Type II = (Also known as cytotoxic hypersensitivity, Effect various organs & tissues)

Type III = (Also known as immune complex hypersensitivity, involve in various organs)

Type IV = (Also known as cell mediated hypersensitivity, Response is against intracellular pathogens)

12. Adverse effects of vaccination?

Followings are the adverse effect in respective kind of vaccination

- Joint pains (Arthritis): Rubella vaccination
- Can be fetal: in case of Pertussis
- Neurological disorders: Influenza vaccination
- DPT vaccination has local effects like Redness, swelling and pain
- Mild/Moderate effects: fever, drowsiness. Vomiting and anorexia
- Severe effects: persistent crying, fever, convulsions, collapse, acute encephalopathy, permanent neurological deficit due to pertussis component.

13. Usage of monoclonal antigen?

The specific monoclonal antibodies can be used against tumor antigen can be served for

- a) As a vehicle for delivering anti-cancer drug to tumor or cancer
- b) Activation of components of innate immune system like complements proteins

14. Combination of VDJ?

The combination of VDJ refers to the recombination of gene segments during the development of T and B cells, contributing to the diversity of antigen receptors.

V (Variable), D (Diversity), and J (Joining) Segments:

- ♣ TCR and immunoglobulin genes undergo VDJ recombination during lymphocyte development.
- ♣ Random combinations of V, D, and J gene segments create diverse antigen receptor structures.
- ♣ This process occurs in the bone marrow for B cells and the thymus for T cells.
- ♣ Results in a vast repertoire of antigen receptors, allowing recognition of a wide range of antigens.

15. How immune system responds against intracellular pathogens?

The secretory immune molecules are ineffective against intracellular pathogens like viruses and intracellular bacteria. The cell mediated immune response is the primary defense against intracellular pathogens. The T-Lymphocytes play vital role in cell mediated immunity. Cell mediated immune response varies according to the residing site of the pathogen

- 1) Cytosolic site (The exogenous pathogens are the pathogens which used to reside in the cytosol of infected cell)
- 2) Vesicular site (The endogenous pathogens are those pathogens which used to reside in the vesicle like phagosomes of infected cell)

16. Adverse effects of inflammation?

Inflammation is associated with general flu like symptoms including

- Fever
 - Chills
 - Fatigue/Lethargy/Loss of energy
 - Headaches
 - Loss of appetite
 - Body pains
 - Muscle stiffness
- 1. Dolor (Pain)
 - 2. Calor (Heat)
 - 3. Rubor (Redness)
 - 4. Tumor (Swelling)
 - 5. Functio laesa (Loss of function)

17. Defects of acute inflammation?

Defects in acute inflammation can lead to various issues:

1. **Insufficient Inflammation:**
 - **Immunodeficiency Disorders:** Weakened immune response (e.g., HIV/AIDS).
 - **Delayed Healing:** Poor wound healing and increased infection risk.
2. **Excessive Inflammation:**
 - **Chronic Inflammation:** Persistent inflammation causing tissue damage (e.g., rheumatoid arthritis).
 - **Autoimmune Disorders:** Immune system attacks own tissues (e.g., lupus).
 - **Hypersensitivity Reactions:** Overreaction to harmless substances (e.g., allergies).
3. **Defective Inflammatory Mediators:**
 - **Cytokine Dysregulation:** Abnormal cytokine production leading to conditions like cytokine storm syndrome.
 - **Defective Chemotaxis:** Impaired immune cell migration.
4. **Defects in Phagocytosis:**
 - **Phagocytic Disorders:** Ineffective pathogen destruction (e.g., chronic granulomatous disease).
5. **Defective Resolution of Inflammation:**
 - **Impaired Clearance:** Prolonged inflammation due to failure in removing dead cells.
 - **Defective Anti-inflammatory Pathways:** Insufficient anti-inflammatory molecule production.
6. **Vascular Defects:**
 - **Abnormal Vascular Permeability:** Inadequate immune cell delivery.
 - **Thrombosis:** Excessive clotting impeding blood flow.

T-Dependent Antigen:

T-dependent antigens are those that do not directly stimulate the production of antibody without the help of T cells

T-independent Antigen:

T-independent antigens are antigens which can directly stimulate the B cells to produce antibody without the requirement for T cell help

18. T-dependent and T-independent antigen?

T-Dependent Antigens

- **Require T helper cells** for a strong immune response.
- **Types:** Proteins or glycoproteins.
- B cells recognize and bind to the antigen.
- Interaction with T helper cells activates and differentiates B cells.
- High-affinity antibodies are produced.
- Immunological memory is generated.

T-Independent Antigens

- **Do not require T helper cells;** directly activate B cells.
- **Types:** Polysaccharides or lipopolysaccharides with repetitive structures.
- B cells are directly activated by the antigen.
- Lower-affinity antibodies are produced.
- Less likely to generate immunological memory.
- **Common in:** Responses to certain bacterial components.

19. Mechanism of activation of B-cell with the independent antigen?

The activation of B cells with T-cell-independent antigens involves direct recognition of antigens by B cells without the need for assistance from T cells. This process typically occurs when the antigens are highly repetitive and can cross-link multiple B cell receptors (BCRs) simultaneously.

i. Recognition of Antigen:

- T-cell-independent antigens are often large and repetitive, such as certain polysaccharides or lipopolysaccharides found on the surface of bacteria.
- B cell receptors (BCRs) on the surface of B cells directly bind to these repetitive epitopes without the need for processing or presentation by antigen-presenting cells.

ii. Cross-Linking of BCRs:

- Binding of the BCRs to the repetitive epitopes leads to cross-linking of multiple BCRs on the B cell surface. This cross-linking is a crucial step for B cell activation

20. What are PAMPs?

PRRs recognize various molecular patterns on the surface of pathogens which are called as ~~pattern~~ associated molecular patterns (PAMPs). These are in the following forms

- Lipopolysaccharides (LPS) (These are recognized by TLRs)
- Flagellins (These are also recognized by TLRs)
- Cell wall component (These are recognized through complement receptor after binding with complement)

21. What is oxygen independent killing mechanism?

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins. Those proteins are bacteriocidal in nature according to their modes of action

22. What is affinity maturation?

A process in the immune system where B cells undergo iterative rounds of mutation and selection to improve the binding strength (affinity) of antibodies to antigens. It occurs primarily during the germinal center reaction in secondary lymphoid organs. Result leads to the production of high-affinity antibodies, enhancing the effectiveness of the immune response over time.

Location: Chromosome 6

Major Loci: DP, DQ, DR

Structure:

Each locus codes for one α -chain and one β -chain.

Both chains form the MHC II complex.

Polymorphic (many alleles), especially DR locus with multiple β chain genes.

Expression: Found on B-lymphocytes and antigen-presenting cells (APCs)

23. Class II MHC?

The class II MHC is encoded by genes for Human MHC which are located at chromosome 6. Structurally, the Class II MHC complex is also composed of three following major loci DP, DQ, DR. The Class II MHC is expressed on B-lymphocytes and antigen presenting cells (APC)

24. Co-stimulatory of B-cell antibodies?

CD40: Interaction with CD40 ligand (CD40L) on T cells provides essential co-stimulatory signals for B cell activation.

B7 Molecules (CD80/CD86): Interaction with CD28 on T cells enhances B cell activation and antibody production.

ICOS (Inducible T-cell Co-stimulator): Interaction with its ligand on B cells promotes T cell-dependent B cell responses.

There is the process of upregulation of co-stimulatory molecule on B-cell which are known as CD40. The engagement of CD 40 occur with its ligand on T-cells which is called as CD40 ligand. For fully activation of Bcells there is also ensured co-stimulatory signaling in B-cells.

BAFF (B-cell Activating Factor): Stimulates B cell survival, proliferation, and antibody production.

APRIL (A Proliferation-Inducing Ligand): Enhances B cell survival and immunoglobulin class-switching.

25. Artificial active Immunization?

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administered to body in less virulent or attenuated form like heat killed forms of microbes.

This form of immunization is achieved by transferring or injecting live, dead or other components of microbes. Followings are the important vaccines which are used as artificial active immunization 1) Live (attenuated) organisms 2) Killed organisms 3) Sub-unit vaccines.

26. Artificial passive immunization?

This form of artificial immunization is the acquired immunization which is achieved by administering gamma globulins or antibodies from other individual's or immune animal's serum. Artificial passive immunity is used against various following acute bacterial infections such as Diphtheria, Tetanus.

27. Explain autoimmunity ?

All mechanisms which are responsible for breakdown of self-tolerance against self-antigens are called as autoimmunity or auto immune reactions.

In this condition, there is generation of immune responses against components of self-tissues. These are harmful or aberrant immune responses against self-tissues. Various products of immune system damage the host tissues like both antibodies and T-cells are involved in autoimmunity. Moreover, genetic predisposition like certain genes of immunoglobulins, TCR & MHC are also associated with various autoimmune diseases. Various environmental factors are also responsible for autoimmunity e.g drugs & infections.

The mechanism of autoimmunity. These are as follows

- Sequestered antigens
- Escape of auto reactive cellular clones

- Lack of regulatory T-cells
- Cross reactive antigens

28. Role of immune system ?

- Defense against infections: AIDS
- Responsive against tissue grafts & newly introduced proteins: Barrier for tissue transplantation & gene therapy
- Surveillance against cancers (Tumors)
- Immune Products e.g Antibodies use in diagnostics & therapeutics

29. What are cytokines?

- Diverse group of non-antibody proteins
- Act as mediators between cells
- Initially these were considered as products of immune cells
- Also considered to act as mediator for immune cells
- Also produced from non-immune cells & mediators for them.
- Used as biological response modifier for treatment of various diseases
- Cytokine is a general term but more specific term used according to their cell of origin

30. Role of Cytotoxic T-Cells?

The cytotoxic T-Lymphocytes are also called as CTLs and CD8+ve. These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot kill the target cells. There is need of differentiation for full activation of CTLs. CTL differentiates from Pre-CTLs in response to following two signals

1. The specific antigen in context of Class I MHC
2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ

31. Which defect are induced due to IgE deficiency?

IgE deficiency is a rare condition where the body doesn't produce enough Immunoglobulin E (IgE), a type of antibody that plays a key role in allergic reactions and parasite infections.

Interestingly, IgE deficiency is not typically associated with increased susceptibility to infections or defects. In fact, people with IgE deficiency may have a lower risk of

developing allergies and allergic diseases, such as asthma, atopic dermatitis, and food allergies.

some research suggests that IgE deficiency might be associated with an increased risk of certain infections, such as: Parasitic infections, Respiratory infections, Autoimmune disorders

32. Enlist effectors function of complement protein.

Secondary or Effector functions: These functions of complement proteins effect on other functions of immune system. These functions of complement proteins are as follows

- Opsonization
- Chemotaxis
- Anaphylaxis

33. Detail of digeorge syndrome

This is the disorder of T-lymphocytes due to congenital thymic aplasia or hypoplasia. This condition is highly immunodeficient as live vaccines also cause infections in the patients suffering from DiGeorge syndrome.

34. Adjuvant and its types?

These are the substances which increase the antigenicity of weak antigens. • Used with vaccine in vaccination • Alum or Aluminium hydroxide • Show undesirable side effects i.e fever

These can be of two forms

- 1) Chemicals
- 2) Biological

35. Antigen, hapten and immunogens ?

Antigen:

An antigen is a substance that can trigger an immune response, causing the body to produce antibodies or immune cells that specifically recognize and bind to it. Antigens

can be proteins, carbohydrates, nucleic acids, or other molecules that are recognized as foreign by the immune system. Examples of antigens include:

- Viral proteins, Bacterial cell walls, Toxins, Pollen, Dust mites

Hapten:

A hapten is a small molecule that cannot trigger an immune response on its own but can bind to a larger molecule (like a protein or carbohydrate) to form a complete antigen.

Haptens are typically small molecules like:

Drugs (e.g., penicillin), Toxins (e.g., urushiol from poison ivy), Metals (e.g., nickel), Chemicals (e.g., formaldehyde)

Haptens become immunogenic (able to trigger an immune response) when attached to a carrier molecule, forming a hapten-carrier complex.

Immunogen:

An immunogen is a substance that can stimulate an immune response, resulting in the production of antibodies or immune cells that specifically recognize and bind to it.

Immunogens are typically large molecules like:

Proteins, Polysaccharides, Viruses, Bacteria

36. Importance of immune system?

The immune system plays a vital role in protecting the body against infections, diseases, and foreign substances. Here are some key importance of the immune system:

1. Protection against infections: The immune system defends against bacteria, viruses, fungi, and other pathogens that can cause infections.

2. Cancer prevention: The immune system can recognize and eliminate cancer cells, preventing tumor growth and development.

3. Tissue repair: The immune system helps repair damaged tissues and promotes healing.

4. Detoxification: The immune system removes toxins and foreign substances from the body.

5. Allergy prevention: The immune system regulates allergic responses, preventing overreactions to harmless substances.

6. Maintenance of homeostasis: The immune system helps maintain the body's internal balance and stability.

37. How cellular immunity protect?

Cellular Immunity: Association of cells for protective functions of immunization

Cellular Immunity Provides protection in following ways

- T-cell immunity: By activating cytotoxic T-cells for killing infected & transformed cells
- By activating macrophages for destroying pathogens
- Activating NK cells for killing transformed & viral infected cells

38. Horizontal and vertical gene transmission?

1) Horizontal transmission This kind of transmission occur usually by person to person contact either sexually or through body fluids like blood and semen

2) Vertical transmission This form of transmission is from mother to fetus during the process of pregnancy, delivery & breast feeding

39. Hypersensitivity I ?

- Also known as Immediate or Anaphylactic reactions
- Involve various tissues • Skin: Urticaria & Eczema • Eyes: Conjunctivitis • Nose: Rhinorrhea, Rhinitis • Lungs: Asthma • GIT: Gastroenteritis
- Symptoms: Minor inconvenience to death
- Usually reaction is quick: 15-30 minutes • Sometimes can be 10-12 hours
- Mediated by IgE
- Primary cells involved: Basophils, Mast cells
- Reaction is amplified by involvement of platelets, Neutrophils & Eosinophils

40. How immune system responds against intracellular pathogens ?

- Secretory immune molecules are ineffective against intracellular pathogens like viruses & intracellular bacteria

- Cell mediated immune response is the primary defense against intracellular pathogens
- T-Lymphocytes play role in cell mediated immunity
- Cell mediated immune response varies according to the residing site of the pathogen

1) Cytosolic site 2) Vesicular site

41. What is difference between artificial and natural immunization?

Natural active immunization

Natural immunization is acquired immunization which is achieved by natural exposure to antigen or acquiring infection by an infectious agent or microorganism e.g measles virus. This is also termed as natural active immunization. This form of natural exposure to antigen stimulate the humoral immunity as a result specific antibodies are generated against exposed microbe. Moreover, there is also activation of cell mediated immunity for intracellular pathogen.

Artificial Active Immunization

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administrated to body in less virulent or attenuated form like heat killed forms of microbes. This artificial exposure with antigen stimulate the humoral immunity in the form antibodies generation. Moreover, artificial activation of cell mediated immunity for intracellular pathogen for example BCG for Mycobacterium tuberculosis. Artificial immunization: acquired immunization by artificial exposure. This artificial exposure to antigen or microbe which is by administration of less virulent (attenuated), heat killed microbes to organism. Artificial exposure to antigen stimulate the humoral immunity for antibodies

42. Cytokine in adaptive immune system:

Followings are the cytokines which act primarily for Adaptive (Specific) immune system

1) IL-2 It is produced by T-helper (Th) cells but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulate B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on T cells in autocrine manner.

2) IL-4 IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE

3) IL-5 IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

4) Tumor Growth Factor (TGF)- β TGF- β is produced by T cells and many other cells types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines

43. Three accessory molecules?

In order to stabilize this interaction, there is the need of additional molecules in addition to TCR & antigen/MHC which are called as accessory molecules.

- **CD4** This molecule of T-cell is for binding to class II MHC which ensures the binding of T-helper (Th) cells to antigen presenting cells
- **CD8** This accessory molecule is for binding to Class I MHC which ensures the binding of cytotoxic T (Tc) cells to target cell
- **CD2** This accessory molecule on the surface of T-cell has a ligand known as Leucocyte Function Antigen 3 (LFA-3) on antigen presenting cells (APC)

44. Antigenic shift and drift?

- **Antigenic shift** This is the major form of change in antigenic structure of antigen as a result the new strains of microbes are generated
- **Antigenic drift** This is the minor form of antigenic variation within the same strain for example influenza (Flu) virus exhibits both antigenic shift and drift for pandemic and seasonal flu respectively. Similarly, bacteria like E.coli, Neisseria gonorrhoeae and Salmonella typhimurium also exhibit antigenic drift.

45. Types of immune system?

Two (02) types of Immunity

Innate Immune system • Non-Specific • First line of defense • Readily available

Adaptive Immune System • Specific • Second line of Defense • Needs to be activated

Innate Immune System	Adaptive Immune System
Response is antigen-independent	Response is antigen-dependent
There is immediate maximal response	There is a lag time between exposure and maximal response
Not antigen-specific	Antigen-specific
Exposure results in no immunologic memory	Exposure results in immunologic memory

46. Characteristics of antibody response ?

- Antibody formation process is well controlled mechanisms with following characteristics
- 1) Self/Non-Self Discrimination • Antibodies would be reactive against non-self antigens
 - Reactivity against self Ag: Autoimmunity
 - 2) Memory • Remembering the nature of antigen • Illicit memory response against same antigen • Anamnestic response • Robust way of clearing antigen • B memory cells
 - 3) Specificity • High degree of specificity of antibodies against specific antigens • Less chances of cross reactivity • Characteristic of adaptive immune response

47. Components of CD3?

CD3 Composed of 5 proteins:CD3 complex

- One Gamma(γ)
- One delta(δ)
- Two Epsilon (ϵ)
- Two Zeta (ζ)

48. Types of Freund adjuvants?

- Freund's complete adjuvant (FCA): water-in-oil emulsion containing antigen with heat killed Mycobacterium tuberculosis
- Stimulate both humoral & Cell mediated immune response for Ab
- Used as booster antigen dose
- Freund's adjuvant is used by mixing with equal parts of antigen

49. Difference between primary and secondary immune response

Primary Immune Response:

- Initial exposure to a pathogen
- Delayed response (4-7 days)
- Limited affinity antibodies
- Short duration (weeks)
- Activation of naive T-cells
- Production of IgM antibodies
- Less effective immune response

Secondary Immune Response:

- Subsequent exposure to the same pathogen
- Rapid response (hours or days)
- High affinity antibodies
- Long duration (months or years)
- Activation of memory T-cells
- Production of IgG antibodies
- More effective immune response

50. Pathology defects of complement system?

- Acute Inflammatory response

- Capillary dilatation
- Exudation of plasma proteins & fluids (Edema)
- Bronchoconstriction
- Mast cells degranulation (Allergic reaction)

51. Difference between primary and secondary antibody response?

Primary Ab response

- First kind of antibody response against antigen
- Major class of antibody is IgM
- Ab are with less affinity & Less persistent

Secondary Ab response

- Second kind of antibody response against antigen
- Major class is IgG but may IgE or IgA but less IgM

How parasite are killed mast cells and eosinophils.

Mast cells and eosinophils work together to fight against parasites:

- Mast cells release cytokines, activating eosinophils.
- Eosinophils release granules, which activate mast cells.
- Together, they create a toxic environment for parasites, leading to their death.

52. Difference between B-cell receptor and T- cells receptors?

B-cells are activated when there is high concentration of antigen for example in case of bacteria the lipopolysacchrides antigens exist in greater quantity. B-cell activation occur after binding of antigen to a surface molecule which is known as **B-cell receptor**.

T-cell receptor:

- Recognize and bind to specific antigens presented by MHC molecules
- Consist of alpha (α) and beta (β) chains with variable (V) and constant (C) regions
- Trigger T-cell activation, proliferation, and differentiation

- Two main types: $\alpha\beta$ TCRs (most common) and $\gamma\delta$ TCRs (recognize non-peptide antigens)

53. Examples of endogenous antigens?

1. Cancer antigens
2. Viral antigens
3. Self-antigens

54. What are the defects appear in IgA deficiency?

IgA deficiency is Commonest form of immunoglobulin deficiency

- Defect in class switching
- Patients are prone to GIT, eye & nasopharyngeal infections
- Diagnosis: measurement of IgA by various immunological methods

55. Briefly explained steps in process of phagocytosis?

- **Recognition:** Phagocyte recognizes foreign particle or microbe.
- **Adhesion:** Phagocyte adheres to the particle or microbe.
- **Engulfment:** Phagocyte extends pseudopodia around the particle or microbe.
- **Ingestion:** Phagocyte closes pseudopodia, forming a phagosome.
- **Fusion:** Phagosome fuses with lysosome, containing digestive enzymes.
- **Digestion:** Enzymes break down the ingested material.
- **Elimination:** Waste products are eliminated through exocytosis.

56. Evidence immune reactivity tumor shown by body?

1. Immune Cell Infiltration:

- **Tumor-Infiltrating Lymphocytes (TILs):** Presence of T cells within tumors.
- **Macrophages and Dendritic Cells:** Other immune cells found in the tumor microenvironment.

2. Elevated Immune Markers:

- **Cytokines and Chemokines:** Increased levels in blood or tumor.

- **Checkpoint Proteins:** Expression of PD-1, PD-L1, and CTLA-4.
- 3. **Tumor-Specific Antibodies:**
 - **Circulating Antibodies:** Antibodies targeting tumor antigens.
 - **Autoantibodies:** Produced against overexpressed tumor proteins.
- 4. **Immune-Related Adverse Events (irAEs):**
 - **Side Effects:** From immunotherapy, indicating immune activation.
- 5. **Tumor Regression and Clinical Responses:**
 - **Spontaneous Regression:** Rare tumor shrinkage without treatment.
 - **Immunotherapy Response:** Tumor reduction from treatments like checkpoint inhibitors.
- 6. **Immunohistochemistry (IHC) and Flow Cytometry:**
 - **IHC:** Staining to identify immune cells in tumors.
 - **Flow Cytometry:** Analysis of immune cell populations.
- 7. **Molecular and Genetic Evidence:**
 - **Neoantigen Load:** High mutational burden producing neoantigens.
 - **Gene Expression Profiles:** Upregulation of immune-related genes.

57. Basic structure of antibodies?

- Composed of following structures

- 1) **Heavy & Light chains:** Four chains, 02 identical light chains, 02 identical heavy chains
- 2) **Disulfide bonds:** Interchain & Intrachain
- 3) **Variable & constant regions:** based on amino acid sequences
- 4) **Hinge region:** makes antibodies flexible to change its shape while performing function
- 5) **Domains:** Folded regions which contain an intra-chain disulfide bond
- 6) **Oligosaccharides:** attached on CH2 domain in most of immunoglobulins

58. Types of antibodies?

T-dependent antigens

- Require T-cells • B-cells are activated with the help of T-cells for antibodies production
- Protein in nature • Contain variety of epitopes with few copies

T-independent antigens

- Not Require T-cells • B-cells are activated directly without the help of T-cells for antibodies production • Polysacchride in nature • Contain same kind of epitopes in polymeric form • More resistant to degradation

59. Neutralization?

The pathogenicity of bacteria is being determined by secreted molecules of bacteria like toxins. These toxins are neutralized by specific antibodies against toxins in a form of antitoxins by the process of neutralization. The binding of antibodies with toxins leads towards immune complex formation. Furthermore, the clearance of immune complex is done by the process of phagocytosis

60. Endocrine/ Paracrine?

- **Paracrine:** effect on nearby cells to cells which secrete cytokine
- **Endocrine:** effect on distant cells through circulation

61. Clonal selection and its principal?

- T-Cell Receptors (TCR)
- Receptor Specificity
- Lymphocyte Repertoire
- After maturation, selection of lymphocytes which are non-reactive to self antigens

Four principles

- 1) Unique Receptor Specificity
- 2) Lymphocyte Activation: Interaction with foreign molecule
- 3) Differentiation of lymphocyte: from single clone
- 4) Selection of lymphocytes bearing receptors against non-self molecules

62. Delayed type hypersensitivity?

Type IV Hypersensitivity : These reactions are also termed as cell mediated hypersensitivity. Due to more time requirement like 48-72 hours, these reactions are also called as delayed type of hypersensitivity. Immunologically, T-cells, Macrophages and monocytes are involved in these reactions.

63. How antibody formed?

A process of immune response against antigen or immunogen

- Can be T-independent or dependent
- B-cells recognize the Ag with its specific receptor
- Transformation of B-cells into plasma cells
- Plasma cells secrete antibodies

64. Cellular barriers?

Cellular barriers are:

- 1) **Neutrophils** • Polymorphonuclear leucocytes (PMNs) • Phagocytose the invading agent & kill intracellularly • Immune cells from the inflamed tissue
- 2) **Macrophages** • Phagocytic cells • Tissue macrophages: Histocytes • Circulating macrophages: Monocytes • Involved in phagocytosis & intracellular killing
- 3) **Natural Killer (NK) Cells** • Have the ability to kill viral infected cells non-specifically • Also kill transformed or Tumorous cells • Role in tumor surveillance
- 4) **Eosinophils** • Blood Granulocytes • Contain granules which are effective against parasitic infections • Also cause the cytotoxicity of parasitic infected cells via receptors non-specifically

65. Explain immune response against transplant?

- **Transplantation:** is the process of moving cells, tissues & organs from one site to another
- One person (donor) to other person (recipient)
- Immune system plays an important role in transplantation
- Act as a barrier for transplantation
- Identifies as foreign and mount an immune response
- Destruction or damage of transplanted tissue
- Transplantation Rejection: Recipient's immune system activation against donor

- Immuno-competent host recognize foreign antigens on grafted tissues
- As a result immune response generated against: graft rejection
- Tissue transplants in immunocompromised individuals leads to immune response by immunocompetent cells in graft
- Host antigens as foreign

66. Secondary lymphoid organs?

Secondary Lymphoid Organs includes

- Lymph Nodes
 - Spleen
 - Tonsils
 - Adenoids
 - Appendix
 - Peyer's patches
- Secondary lymphoid organs, including lymph nodes, spleen, and mucosal associated lymphoid tissues (MALT) such as gut-associated lymphoid tissue (GALT), trap antigens and provide sites for mature lymphocytes to interact with these antigens.

67. Name the drug that induce hemolytic anemia?

The examples of exogenous antigens include Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia.

68. Complement fixation test?

- Ag & Ab reaction which can be detected on the basis of ability to fix or consume complement
- Used for determining good complement fixing antibodies e.g IgG or IgM
- Unknown Ag : Known Ab
- Complement proteins
- Presensitized RBC with anti RBC antibodies
- Lysis of RBC: No Complement fixation by Ag & Ab complex: No Ag: Negative CFT
- No Lysis: Complement fixation by AG & Ab complex: Ag is present: Positive CFT

- CFT is used for determining Gonococcal Ab in serum

69. Signs of inflammation?

Signs of inflammation: pain, heat, redness, swelling & loss of tissue functions.

Inflammation can be beneficial for host in order to remold the damaged tissue •

Undesirable effects: in the form of signs of inflammation • Dolor (Pain) • Calor (Heat) • Rubor (Redness) • Tumor (Swelling) • Functio laesa (Loss of function)

70. What is direct and indirect ELISA?

Direct ELISA • Unknown Ab can be determined • Known Ag is coated on solid phase (Glass surface) • Serum containing unknown Ab is added • Another ab which is precoated with enzyme e.g horseradish peroxidase • Substrate: H₂O₂

Indirect (Sandwich) ELISA • Unknown Ag can be determined • Known Ab is coated on solid phase (Glass surface) • Serum containing unknown Ag is added • Another Ab which is precoated with enzyme e.g horseradish peroxidase • Substrate: H₂O₂

71. Onco fetal antigen?

The tumor antigens which are also expressed by normal cells. These are expressed in high levels by tumor cells as compare to normal cells. These kind of antigens are also called as onco-fetal antigens as these are expressed during early development & lost during adult life.

There are following two types of onco-fetal antigens

- 1) Alpha-fetoproteins (AFP)
- 2) Carcino-embryonic antigens (CEA)